

Shell, Biological Formats, and Quality Control

Bioinformatic tools

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Quick poll: pe.app/cmaslard1

One question from Session 1 before we move on.

You are handed sequencing reads from an experiment that has been running for decades. Before anyone trusts a single result from them: are these reads good enough to use?

From FASTA to FASTQ

FASTA stored a sequence and its name.

FASTQ adds: how much the sequencer trusts each base.

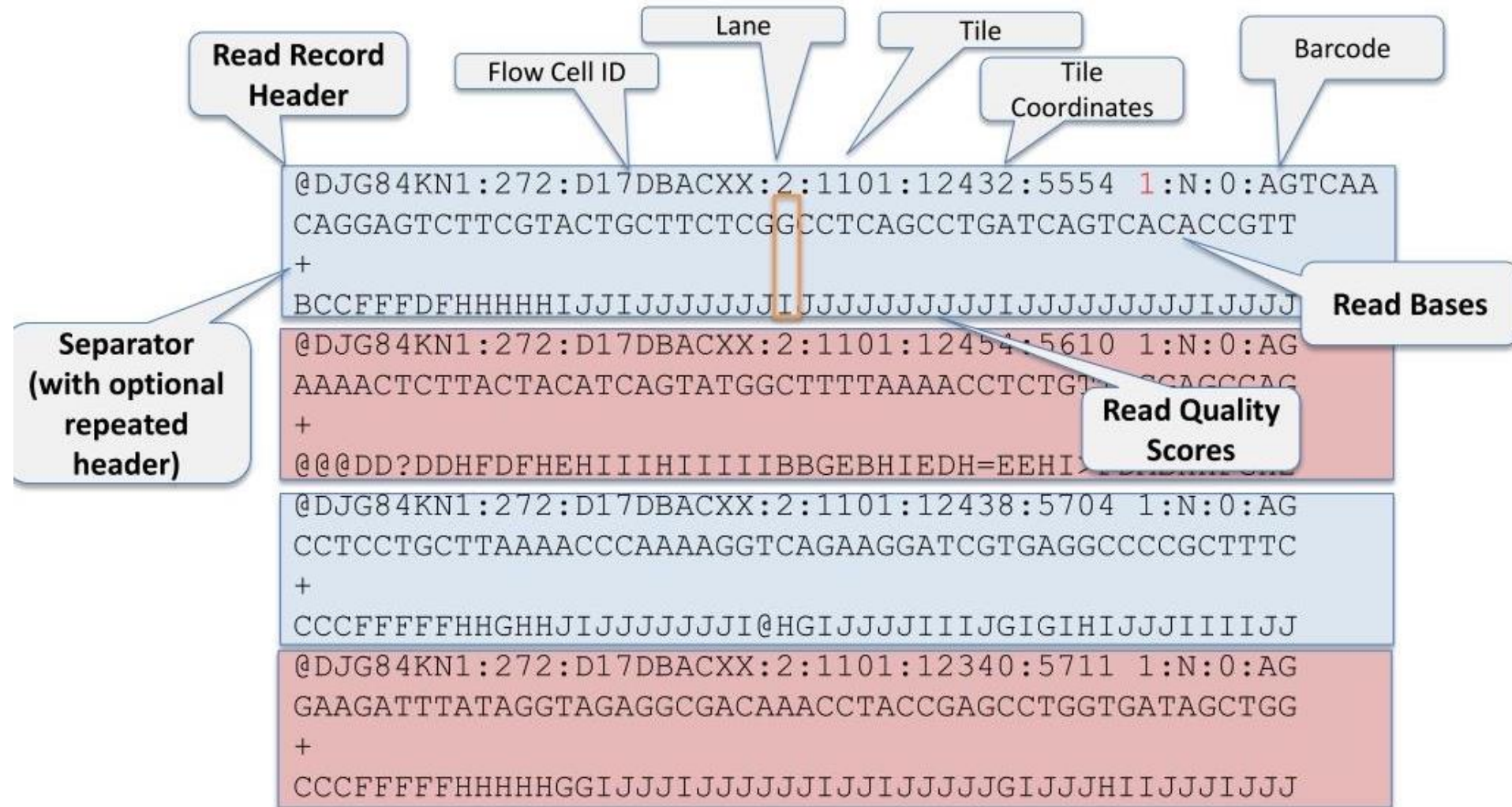
Every FASTQ record = exactly 4 lines.

The diagram shows four lines representing a single FASTQ record:

- Line 1: Sample name and read info (`@SRR098026.177 HWUSI-EAS1599_1:2:1:1:2025 length=35`)
- Line 2: DNA sequence (`CNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN`)
- Line 3: Separator character (`+`)
- Line 4: PHRED quality score (`#!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!`)

A box highlights the alignment between the 10th base of the DNA sequence ('C') and its corresponding quality score ('!'). Arrows point from labels to their respective parts: "Sample name and read info" points to Line 1; "DNA sequence" points to Line 2; and "PHRED quality score" points to Line 4.

FASTQ Format (Illumina Example)



NOTE: for paired-end runs, there is a second file with one-to-one corresponding headers and reads.

What a quality score actually is

FASTA stored a sequence and its name.
FASTQ adds: how much the sequencer trusts each base.
Every FASTQ record = exactly 4 lines.

```
@SRR098281.1 length=36
GATTGGGGTTCAAAGCAGTATCGATCAAATAGTAA
+
IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII
```

Each character on line 4 = a Phred quality score for that one base. Higher = more confident.

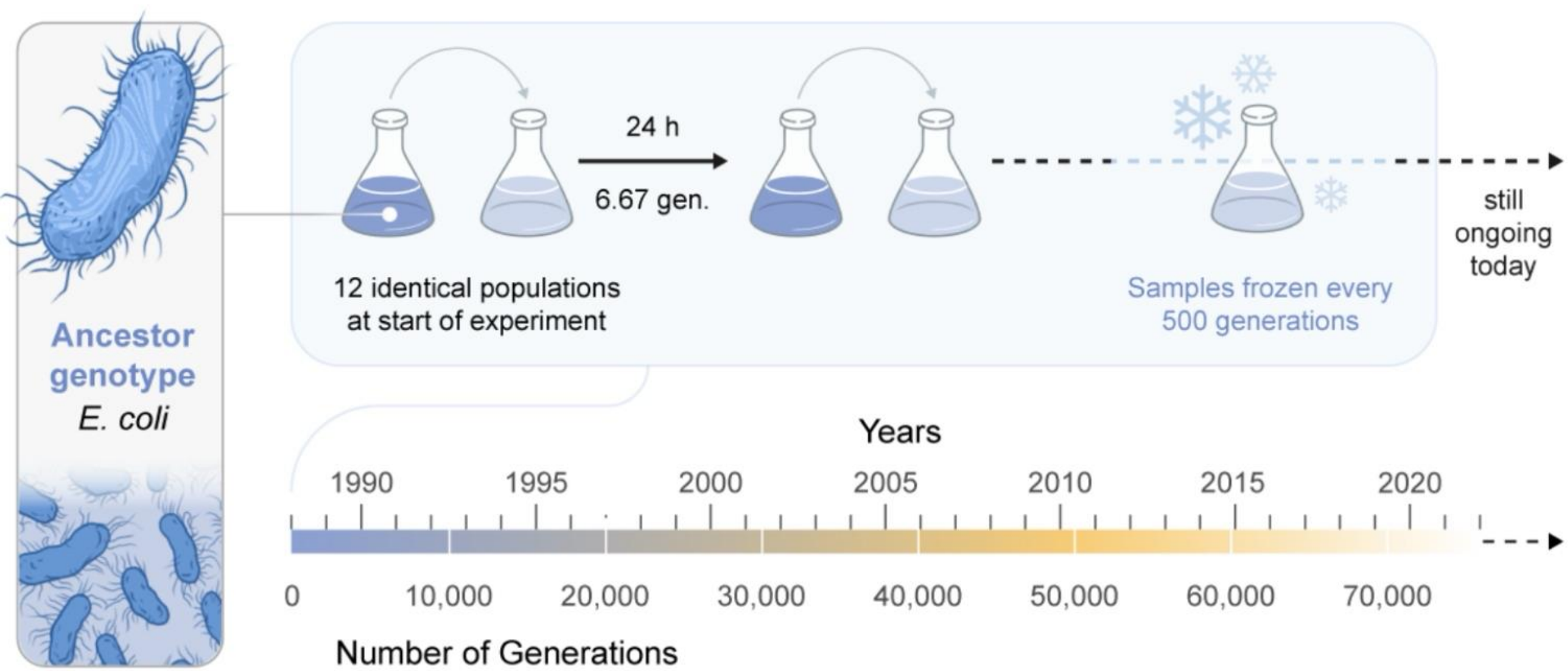
Character	Score	Meaning
!	0	No confidence
#	2	very likely wrong
,	11	doubtful
:	25	acceptable
F	37	good
I	40	Excellent

Why a tool, not your eyes

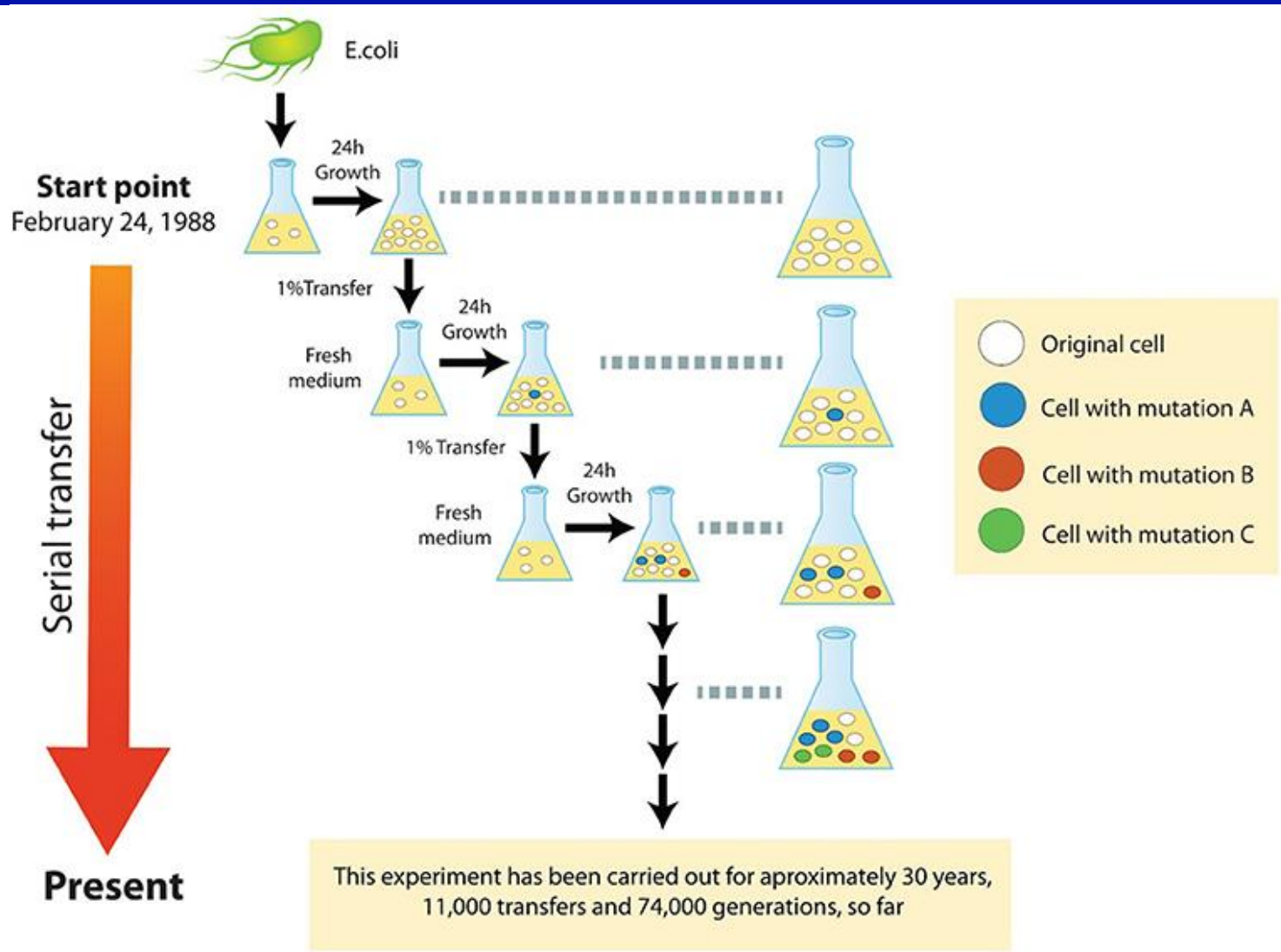
A real run = millions of reads. FastQC decodes every quality line, plots score by position, across the whole file, in one report.



Worked example: 50,000 generations of E. coli. Lenski's Long-Term Evolution Experiment (LTEE), since 1988 !!!



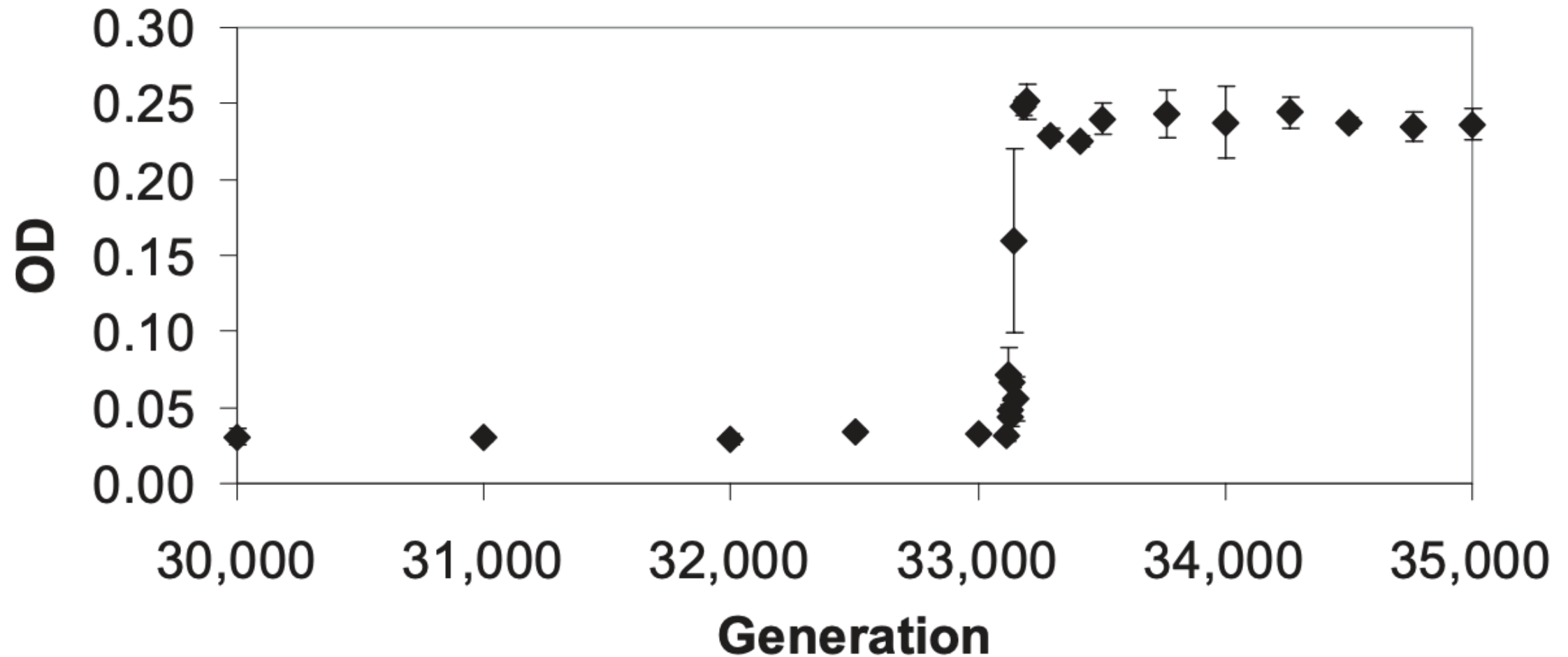
Worked example: 50,000 generations of E. coli. Lenski's Long-Term Evolution Experiment (LTEE), since 1988 !!!



The “plot twist” of citrate

- *E. coli* grew mainly on **glucose**
- The medium also contained **plenty of citrate**
- But *E. coli* **couldn't use citrate** as a carbon source under these conditions
- **For >30,000 generations, nothing changed...**

~31,500 generations later... something changed.



- ~31,500 →  Cit⁺ appears in Ara-3

We have sequencing reads from this experiment.

**But before asking what evolved, we need to ask something much more boring - and essential:
Can we trust the reads?**

Real reads from three timepoints: 5,000 / 15,000 / 50,000 generations. Question for today: are these reads trustworthy? (Not today's question: what actually changed in the DNA, that's the bonus / later sessions.)

What "quality doctor" means today

1. Fetch 3 FASTQ files, look inside one by hand.
2. Run FastQC on all three.
3. Build a checklist from a known good/bad example pair.
4. Diagnose your own two extreme timepoints.
5. Individually: decide trust/flag on an unseen sample, with two named observations.

New:

`mv, module load, fastqc`

Still yours from Session 1:

`pwd, ls, mkdir, cd, cp, head, tail, wc -l, grep`